# InView IIoT Platform

INIT Technologies

## Who We Are

INDAS Group / INIT Technologies — Novi Sad, Serbia

- INDAS Group founded **1992** — 30+ years in industrial software
- INIT Technologies spun off **2020** — focused on software product R&D
- **inVIEW** first commercial release: **2011**, microservices + cloud-native since 2017
- Full-stack capability: field protocols → edge → cloud → AI/ML → web/mobile
- In-house R&D — no outsourcing, continuous development

## The Industrial Data Problem

Every factory floor generates massive amounts of data. Most of it never gets used.

- PLCs, sensors, smart meters, IoT devices — all different vendors, all different protocols
- OT (Operational Technology) and IT live in separate worlds — no shared data layer
- Production events recorded **manually on paper** — shift logs, downtime reports, quality checks
- Multiple disconnected software tools — one for planning, another for reporting, another for maintenance
- Data moved **by hand between systems** — Excel exports, manual re-entry, email attachments
- No single source of truth — decisions made on yesterday's numbers, not live data
- Critical machine events go unnoticed until it's too late — no real-time alerting on the shop floor
- Historical data locked in silos — impossible to correlate machine behavior with production output

# Introducing inVIEW

**Cloud-driven. Edge automation. AI-enhanced.**

inVIEW is an **Industrial IoT platform** that connects your machines, devices, databases and software — and lets you build real-time dashboards, SCADA visualizations, and MES workflows.

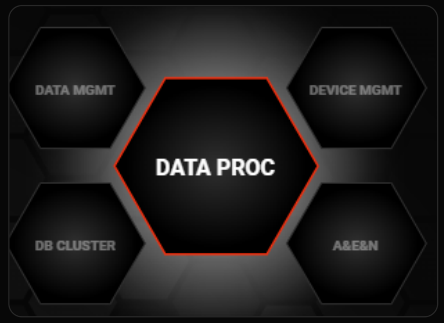
Three core promises:

- **See everything** — real-time data from every device, anywhere
- **Control everything** — remote commands, automation, alarms
- **Understand everything** — AI/ML that finds patterns you can't see manually

> *"It's not just monitoring. It's an evolution of how you run your production."*

THE PLATFORM — 6 MODULES

## CORE ENGINE — Real-Time SCADA at the Center




The heart of the platform.

- Real world data inside the Platform
- Live variable streaming via **WebSocket** — data appears the moment it changes
- Multi-source data aggregation — heterogeneous devices, one unified view
- **Remote control** of field devices — write commands from anywhere in the world
- **Alarm management** — configurable thresholds, deadband, double-check validation, false alarm prevention
- Multi-channel notifications: SMS, Email, Push
- Complete alarm and event history — full audit trail

> *"One platform. Your whole production floor. Live."*

# DEVICES — Connect Anything

**We speak the language of your machines.**

*Natively supported protocols:*

- **Siemens S7** — direct PLC connection
- **OPC UA / OPC DA** — industrial interoperability standard
- **Modbus TCP** — most common industrial protocol
- **IEC 60870-5-104** — power grid, utilities, telecontrol
- **MQTT / LoRaWAN** — modern IoT sensors
- **i3x** — inVIEW proprietary protocol for gateway communication

*inVIEW IoT Gateways — for everything else:*

- Hardware gateways that bridge legacy equipment to the platform
- Support: **Modbus RTU/TCP, Siemens S7, Ethernet/IP, Profinet, OPC**
- **Custom script execution on the gateway** — JavaScript logic runs locally, at the source
- **Store & Forward** — buffers data locally during connectivity loss, syncs automatically
- Gateway software is open for custom driver development

> *"If your machine generates data, we can connect to it — and process it right where it lives."*

## NO-CODE IDE — Your Team Builds It

The people who know your process — build the application.

- **Drag-and-drop screen builder** — no programming required
- Configurator runs entirely in a browser — no software to install
- Your technician builds a new screen, not your software developer

One tool — web and mobile:

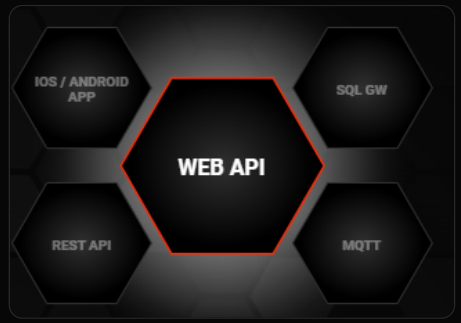
- Screens built in the IDE render on **web browser** and in the **native mobile app** (iOS & Android)
- No separate mobile development — same screen definition, same data, same configuration

What this means in practice:

- Not dependent on a vendor for every configuration change
- IT team or production engineers own the setup
- **Templates** — configure one machine, replicate to 50 identical ones instantly
- New production line → new screen in days, not months

> *"Most SCADA vendors charge you every time you want to change a label. Your team does it themselves — and it works on every device."*

# ENDPOINTS — Connect to Everything Outside




**inVIEW doesn't live in isolation. It talks to your entire ecosystem.**

**InView visualisation tools**

- **Web Client** — expose live and historical tag data via standard InView Client — any browser, app
- **Mobile App** — native iOS & Android app — same screens built in the No-Code IDE, always in sync, works on the shop floor
- **Power BI** — connect inVIEW as a data source for business-level reporting, KPI dashboards, and management overviews
- **Grafana** — plug inVIEW into Grafana for open-source analytical dashboards and time-series visualization

**External systems to integrate with via**

- **REST API** — standard HTTP endpoints for any external system
- **MQTT broker** — publish data to any MQTT subscriber
- **Webhooks / Event Hooks** — trigger external actions on alarms or data events

> *"Any system that can send or receive HTTP, WebSocket, or MQTT can integrate with inVIEW."*

## AI / ML — Your System Gets Smarter Over Time

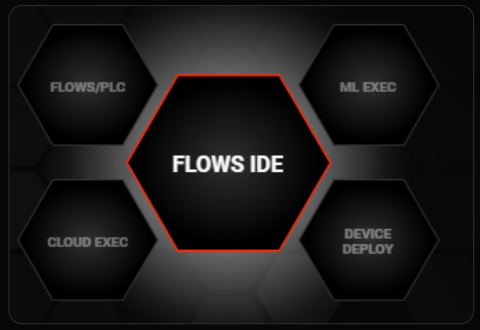
**Built in. Not bolted on.**

What the AI layer does:

- **Anomaly Detection** — Z-Score, IQR, Rate-of-Change, Timeout — ensemble detection for accuracy
- **Self-Prediction** — LSTM neural networks + Random Forest for time-series forecasting
- **Feature Selection** — Pearson, Spearman, Mutual Information — finds which variables actually matter
- **Model Versioning** — incremental model updates, no full retraining needed
- **MLflow integration** — professional ML lifecycle management

> *"Your press temperature starts drifting at 02:00 AM. By 06:00 AM it would have caused a quality defect. The system flags it at 02:15 AM. Your night shift fixes it before morning. That's pattern recognition on your own historical data."*

# EDGE — Intelligence Where the Data Is Born




**Processing logic runs directly on the edge device — at the machine, at the site.**

What edge computing means here:

- Automation scripts run **locally** — no cloud round-trip for time-sensitive decisions
- Works fully **offline** — no internet connection required for local operations
- Reduces bandwidth — only relevant, processed data goes upstream
- Latency-sensitive control stays on-site

Edge + Gateway combination:

- inVIEW Gateway runs edge scripts at the protocol bridge level
- Custom JavaScript logic executes at the source
- Local alarm detection, filtering, and aggregation before cloud sync
- Scales to hundreds of distributed edge nodes from one central management interface

> *"You have a press line with poor connectivity. With Edge, that line is fully autonomous — it monitors, reacts, logs — with or without a live cloud connection."*

## Deployment — Your Rules, Your Infrastructure

**Same platform. Same features. Your choice of where it runs.**

| Model | What it means |
|---|---|
| Cloud (SaaS) | We host and manage — you just log in |
| On-Premise | Fully on your servers, your data center, your control |
| Hybrid | Some on-site, some cloud — best of both worlds |
| Edge + Cloud | Local autonomy + cloud visibility and analytics |

**No feature tiering by deployment model.**

> *"You don't get a stripped-down version for on-premise. Same features, same platform — wherever you run it."*

## Proven at Scale

Not a demo system. Production numbers.

- **100,000 simultaneous screen views** — live, on microservices architecture
- Multi-tenant: multiple enterprise customers on shared infrastructure
- Apache Kafka event streaming — 17 system-wide topics, built for throughput
- Redis caching — sub-millisecond data access
- Kubernetes orchestration — horizontal scaling, zero-downtime deployments

What this means for you:

- Start with one production line. Scale to your entire facility. Same platform.
- Add MES, add AI — no platform migration needed.

> *"We built it right from the beginning. The architecture assumptions from 2017 have held up at scale we didn't fully anticipate."*

## Integration — We Connect to Your World

**inVIEW doesn't replace your existing systems. It integrates with them.**

- **SAP** bidirectional — production orders flow in, execution data flows back automatically
- **ERP / MES systems** — standard REST API integration with any modern platform
- **Historians and databases** — read from existing data stores
- **Pre-built connectors** — common integration scenarios ready out of the box

The SAP story in practice:

- SAP sends production order → visible on floor screen immediately
- Operator executes: quantities, materials, downtime all tracked
- At order close → actual data confirmed back to SAP automatically
- No manual data entry. No phone calls. SAP always reflects reality.

> *"The gap between what SAP thinks happened and what actually happened on the floor — that gap closes completely."*

## Fast Implementation & Real Support

Two things that matter after you sign.

Fast implementation:

- Projects go live in **weeks**, not years
- No-code configuration dramatically reduces setup time
- Templates from similar deployments accelerate new projects
- Local team — no outsourcing, no timezone issues

High-level support:

- Direct access to the people who built the system
- No tier-1 / tier-2 / ticket queues
- We know your specific installation
- Reaction time measured in hours, not days

> *"A competitor lost a deal to us once. 18 months later the customer called us — their implementation still wasn't live. We were in production in 3 weeks."*

## Why inVIEW — The Short Version

If you remember one thing from today:

- ✅ **Proven** — 15+ years in real production environments, active development
- ✅ **Open** — connects to everything you already have, no rip-and-replace
- ✅ **Fast** — weeks to go live, not years
- ✅ **Configurable** — your team owns it, not the vendor
- ✅ **Complete** — SCADA + MES + AI/ML in one platform
- ✅ **Supported** — directly by the people who built it

> *"You don't need to rip anything out. You need a platform that makes your existing setup smarter."*

## Tarkett — Same Industry, Same Challenges

**Global flooring leader. Similar production reality to yours.**

- Tarkett Backa Palanka: **PVC, Wood, and Textile production**
- High-volume continuous manufacturing — multiple product lines running in parallel
- Complex machinery, tight quality tolerances, high cost of downtime
- **SAP** as the ERP backbone of the entire operation
- INIT partnership running since **2016** — two divisions, still expanding

> *"Flooring. Wood panels. Continuous production lines. Quality-critical. SAP-integrated."*

# Tarkett Wood — The Challenge

**Wood panel production. Complex. Multi-stage. SAP-connected.**

**Before inVIEW MES:**

- Production was planned in SAP — but execution on the floor was disconnected
- No real-time work order tracking — operators worked from printed sheets
- Downtime and stoppages recorded manually — hours later, sometimes not at all
- Shift reports took **45 minutes** to complete manually — every single shift
- Quality issues not traceable back to specific production parameters
- SAP always lagged reality — no one trusted the numbers

> *"Management could see what was planned. Nobody could see what actually happened."*

## Tarkett Wood — SCADA + MES (since 2021)

A full Manufacturing Execution System built on the inVIEW SCADA foundation.

SCADA layer — real-time visibility:

- Full production floor visualization — presses, brush lines, packaging, all zones live
- Real-time machine status, cycle parameters, trend analysis
- TV screens on the shop floor — always-on production overview for operators

MES layer — planning and execution:

- **Production planning** — visual planning board, shift scheduling, line assignments
- **Work order lifecycle** — SAP order → floor screen → execution tracked step by step
- **Cycle evidence** — every press cycle recorded: duration, parameters, outcome
- **Downtime management** — stoppages auto-detected, reason codes logged, duration tracked
- **Quality & yield** — per-order data, reject tracking, product version management
- **Shift reports** — generated automatically at shift end. Zero manual input.

## SAP Integration — Data Flows Both Ways

**inVIEW and SAP talk to each other — automatically.**

**inVIEW → SAP (data flowing back to ERP):**

- Actual production quantities confirmed per order
- Material consumption reported
- Downtime events and stoppage reasons
- Work order status updated in real time
- Quality deviations flagged

**SAP → inVIEW (data flowing down to the floor):**

- Production orders pushed directly to floor screens
- Material requirements, BOMs, product versions
- Shift schedules and planning parameters

> *"SAP always reflects what actually happened on the floor. Not because someone entered data — because the system entered it automatically."*

## Tarkett — Results

Wood Division (MES since 2021):

- Production planning fully digitalized and connected to SAP
- Every work order tracked from SAP creation to floor completion
- Every press cycle recorded — full production traceability
- Downtime automatically detected and categorized
- **Shift reports automated** — 45 minutes of manual work → 0
- Live quality data per order, per shift, per product version

PVC Division *(running since 2016 — mentioned for reference):*

- Full SCADA monitoring and control across all production zones
- OEE tracking, command system, SAP-integrated production orders

> *"They started with one line in 2016. Today they run two full divisions on inVIEW. That's 10 years of continuous expansion — the real proof isn't the sale, it's that they keep growing it."*

TARKETT WOOD — SCADA LIVE VIEW

TARKETT WOOD — DAILY PRODUCTION EVIDENCE

| RADNA STANICA | A KLASA | DORADA MD | DORADA RD | ŠKART | GRAND TOTAL |
|---|---|---|---|---|---|
| PBP400 - Presa | 6821.45M2 | 0M2 | 0M2 | 36.18M2 | 6857.63M2 |

| MATERIJAL | ISPUNJENOST | PLAN A KLASE | A KLASA | DORADA MD | DORADA RD | ŠKART | GRAND TOTAL | ISPUNJENOST PRVE KLASE |
|---|---|---|---|---|---|---|---|---|
| 940099235 - PRE TIM13-OAKOR200x2289-CT87-WAVE | 36.4% | 2453.28M2 | 893.1M2 | 0M2 | 0M2 | 0M2 | 893.1M2 | 100% |
| 940257417 - PRE ALG14-OAKNA200x2289-B3-NT96 | 100.95% | 896.07M2 | 904.55M2 | 0M2 | 0M2 | 0M2 | 904.55M2 | 100% |
| 940317970 - PRE SAL14-OAKOR200x2289-B2-PT96 | 89.56% | 5609.6M2 | 5023.8M2 | 0M2 | 0M2 | 36.18M2 | 5059.98M2 | 99.28% |

| | | | | | |
|---|---|---|---|---|---|
| PBP420 - Braširanje | 0M2 | 0M2 | 0M2 | 0M2 | 0M2 |
| PBP450 - Braširanje Četkanje | 5936.59M2 | 0M2 | 0M2 | 0M2 | 5936.59M2 |
| PBP455 - Digitalna stampa | 0M2 | 0M2 | 0M2 | 0M2 | 0M2 |
| PBP456 - Novi digi | 0M2 | 0M2 | 0M2 | 0M2 | 0M2 |
| PBP465 - Rucni bevel | 0M2 | 0M2 | 0M2 | 0M2 | 0M2 |
| PBP470 - Novi homag | 3549.43M2 | 0M2 | 0M2 | 0M2 | 3549.43M2 |
| PBP480 - BKS | 0M2 | 0M2 | 0M2 | 0M2 | 0M2 |
| PBP490 - Uljenje | 0M2 | 0M2 | 0M2 | 0M2 | 0M2 |
| PBP500 - Lakiranje | 2406.06M2 | 0M2 | 0M2 | 0M2 | 2406.06M2 |
| PBP550 - Rucna dorada | 0M2 | 0M2 | 0M2 | 0M2 | 0M2 |
| PBP600 - Stari homage | 3523.13M2 | 0M2 | 0M2 | 0M2 | 3523.13M2 |

* u koloni trajanje vrednosti prikazane u zagradi, odnose se na trajanje naloga bez prekida sa pomnozenim OEE

## PRS1 - 52h30min

| NALOG | MPS PLAN | MATERIJAL | RADNA STANICA | KOLIČINA (M2) | MOGUĆI POČETAK PO PRETHODNOM | PLANIRANI POČETAK | PLANIRANI KRAJ | TRAJANJE | MIN VREME ČEKANJA | NERADNI PERIODI |
|---|---|---|---|---|---|---|---|---|---|---|
| 22471 - 940321552 - PRE EUP13-OAKREG200x2289-B2-WENT87-MOOS | W08_2026 | 550233026 | PBP400 => PBP400 | 296.46 | | 19.02.2026. 07:51 | 19.02.2026. 09:06 | 1h15min (1h15min) | 0h | |
| BUD MPS - 41179 - 940337350 - PRE BLR14-OAKURO171x2444-B2-WENTH96-SIED | W08_2026 | 550035073 | PBP400 => PBP400 | 851.12 | | 19.02.2026. 09:06 | 19.02.2026. 13:37 | 4h31min (4h31min) | 0h | |
| BUD MPS - 41180 - 940337372 - PRE BLR14-OAKURO171x2044-B2-WENTH96-SIED | W08_2026 | 550035074 | PBP400 => PBP400 | 1040.33 | | 19.02.2026. 13:37 | 19.02.2026. 19:51 | 6h14min (6h14min) | 0h | |
| BUD MPS - 41182 - 940334916 - PRE SHA13-OAKRO144x1865-E1-EGN5-$82-EGRE | W08_2026 | 41016002 | PBP400 => PBP400 | 2158.81 | | 19.02.2026. 19:51 | 20.02.2026. 11:03 | 15h12min (15h12min) | 0h | |
| BUD MPS - 41183 - 940334872 - PRE PRF13-OAKURR144x1865-E1-RWN5-$82-CRE | W08_2026 | 41016011 | PBP400 => PBP400 | 1057.62 | | 20.02.2026. 11:03 | 20.02.2026. 18:29 | 7h26min (7h26min) | 0h | |
| BUD MPS - 41188 - 940335178 - PRE HAR13-OAKRP199x2015-E1-ESN5-$82-NORD | W08_2026 | 7879031 | PBP400 => PBP400 | 1741.08 | | 20.02.2026. 18:29 | 21.02.2026. 02:39 | 8h10min (8h10min) | 0h | |
| BUD MPS - 41194 - 940335004 - PRE SHA14-OAKRO171x2015-E1-ATN5-$82-SGRE | W08_2026 | 7876088 | PBP400 => PBP400 | 1584.62 | | 21.02.2026. 02:39 | 21.02.2026. 12:21 | 9h42min (9h42min) | 0h | |

## BRSH - 92h14min

| NALOG | MPS PLAN | MATERIJAL | RADNA STANICA | KOLIČINA (M2) | MOGUĆI POČETAK PO PRETHODNOM | PLANIRANI POČETAK | PLANIRANI KRAJ | TRAJANJE | MIN VREME ČEKANJA | NERADNI PERIODI |
|---|---|---|---|---|---|---|---|---|---|---|
| 22511 - 940321553 - BRA EUP13-OAKREG200x2289-B2-WENT-MOOS | W08_2026 | 550233026 | PBP450 => PBP450 | 858.75 | | 19.02.2026. 09:06 | 19.02.2026. 11:54 | 2h48min (2h48min) | 0h | |
| BUD MPS - 41179 - 940337351 - BRA BLR14-OAKURO171x2444-B2-WENTH-Y-SIED | W08_2026 | 550035073 | PBP400 => PBP450 | 849.42 | 19.02.2026. 13:38 | 19.02.2026. 13:38 | 19.02.2026. 16:40 | 3h2min (3h2min) | 0h | |
| BUD MPS - 41180 - 940337373 - BRA BLR14-OAKURO171x2044-B2-WENTH-Y-SIED | W08_2026 | 550035074 | PBP400 => PBP450 | 1038.25 | 19.02.2026. 19:51 | 19.02.2026. 19:51 | 20.02.2026. 00:17 | 4h26min (4h26min) | 0h | |
| BUD MPS - 41182 - 940334917 - BRA SHA13-OAKRO144x1865-E1-EGN5-Z$-EGREY | W08_2026 | 41016002 | PBP400 => PBP450 | 2158.81 | 20.02.2026. 11:03 | 20.02.2026. 11:03 | 21.02.2026. 00:12 | 13h9min (13h9min) | 0h | |

## Validacija ciklusa

**Vreme završetka radnog ciklusa:**

15. 2. 2026., 22:00:00

| | | | |
|---|---|---|---|
| Trajanje ciklusa | 480.00 min | 14:00 - 22:00 | |
| Vreme rada | 66.00 min | Brzina proizvodnje | 4235 $m^2$/8h |
| | | A klasa | 582.32 $m^2$ |
| | | RD | 0 $m^2$ |
| | | MD | 0 $m^2$ |
| | | Škart | 0 $m^2$ |
| | | Ukupno | 582.32 $m^2$ |
| Zastoji | 0.00 min | | |
| Mikrozastoj | **414.00 min** | **86.25 %** | |

**Mikrozastoji su 86.25% što je veće od 5.00%**

Knjiženje | Zastoj

## Evidentirana A klasa

## What Could This Look Like for Parador?

**Starting point — a conversation, not a commitment.**

Typical first step with a manufacturer like you:

- Short technical workshop — we look at your machines, protocols, data sources
- Define scope: which line, which KPIs, which integrations matter most
- Pilot deployment — one line, live in weeks
- Expand based on results

What you get from day one:

- Live production visibility on the line you start with
- Alarm management — know before things fail
- Foundation for MES and SAP integration when you're ready

> *"We don't ask you to bet the whole factory on us. Start where the pain is biggest. Let the results speak."*

# Thank You

**inVIEW IIoT Platform — INIT Technologies**

*The platform that connects your machines, your people, and your data.*

**Next step:** A short technical workshop at your site — we look at your current setup and define what a pilot could look like.

Robert Sabo | INIT Technologies
robert@industrial-it.software | inview.software

> *"You already have the data. Let's make it work for you."*